My Teaching Philosophy

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My approach to teaching is largely guided by the simple fact that the teacher cannot exist without the student. If one shows up with the purpose of delivering teachings, but the audience is not ready or willing to receive it, then teaching cannot happen. In this way, teaching exists as a sort of harmony or play between student and teacher. As in a game, if I get too caught up in my own movements or strategy, then I miss crucial information or perspective from my opposition. To push the play analogy a little further, treating students as an abstract group to whom I deliver teachings is like planning out all my moves at a chess board before sitting down to play. While seemingly prudent, any attempt to actually do this quickly reveals this is pure nonsense. Students, then, are not an abstract group to whom I deliver teachings. Rather, they are made up of individual human beings who experience awe, curiosity, and beauty as well as boredom, fatigue, and distraction. For this reason, my approach to teaching centers around the perspective of the student.

Reflecting on my experience as a student, I find several common themes underlying times I was engaged with a course and instructor, and times when I was not. Here are a few of these observations, and how they guide my teaching currently.

Respect for the student as an individual. When teachers treated me with respect me as an individual, I was way more willing to pay attention to the course. There were several times where a teacher or professor engaged with my specific, but perhaps tangential questions or situations. These not only put me at ease and provided a sense of belonging, but have also fueled my interest and curiosity with the rest of the course. Being seen and respected as a human seems like a basic starting point in a harmony between two people. Conversely, I've actively refused to engage in classes where the instructor treats me like I'm not worth their time, either by dismissing my curiosities or treating me as a machine or number. These experiences have lead me to treat every individual who is willing to show up with the dignity and respect they deserve as humans. During my lessons, I always actively engage with every question, even if that means having to email the student individually afterwards for a more thorough response. I have also went out of my way to create additional challenging problem sets for the more advanced students who appear bored, and have scheduled additional time and instruction for those students who are struggling.

Information is cheap, but humanity is valuable. I rapidly stopped showing up for lectures when it was clear that the instructor's only goal was to deliver information. Whether this was from reading off of prepared slides, or powering through endless definitions, examples, theorems, and proofs, these experiences always left me with the feeling that I could get this information equally well from a textbook. Especially in this age of AI, information has become incredibly cheap. The concepts of most courses are available to anyone anywhere at the push of a couple of buttons. These questions often sat in the back of my mind when I had to decide whether to make a commute to class or not. The more classes were conveyor belts of information, the less likely I was to show up. Lectures and instructors must therefore, in my mind, provide value that goes beyond information if they want students to show up. I believe that this value is delivered in the *humanity* of the subject matter. For example, I find lessons rich with meaning when the instructor can share their feelings, intuition, or organic approaches to concepts or problems, as these are often far more understandable or relatable to my own experience of the world. These things bring the subject to life, rather than having it be dry and abstract. In my lectures and lessons, I make a deliberate effort to begin the discussion with a connection to my audience's experience. Every definition and result to me is a refinement of thinking that becomes natural to come up with on ones own given the right setting. An ideal that I often strive for in lecturing is to have my audience come up with an idea before I say it.

The struggle is not a bug, it's a feature. I have several vivid memories of learning that came from the struggle of trying to solve difficult problems. I would sometimes try a single problem for days at a time; many different approaches, getting stuck, and then coming at it again later. When I finally had a stroke of insight to solve the problem, it was not only incredibly satisfying, but also taught me to understand the

material in a deep and lasting way. I strongly feel that this sense of struggle and participation in learning is crucial to a rich understanding. Drilling computational exercises may build a sort of muscle memory, but its in connecting ideas together and stepping just outside the material that really helps put everything in perspective and yields a sharper intuition. This is, though, not an easy balance to get right. A problem that is too easy may leave the student feeling bereft of meaning, while one that's too hard or complicated may lead one to give up. My ideal assignments would be split up into sections. At the base level, the student can practice basic calculations relating to the material. A next layer up, the student might have to couple those calculations with an idea from class to peer deeper into *meaning*. Lastly, I would always include a section for the student to go beyond what was taught by connecting several ideas together, or even coming up with a novel approach of their own.

Going forward. Throughout my years of teaching and tutoring mathematics, I've learned that there's always a better way to either explain something, or lead someone to a discovery. This journey of mine to empower students through education in mathematics knows no end, and my skills can always be further refined. Some specific goals I have are largely around lecture design. In particular, I would like to create an environment in my lectures where my students can comfortably see me struggle or make mistakes while solving problems, to see the human side of doing mathematics. In addition, I want to push myself to simplify every lecture, leaving strategic holes for students' curiosities to take root, as well as having one simple big picture idea for every lecture.

A reflection of my performance teaching is always visible right in front of my eyes every moment that I'm teaching, in the faces and postures of the students. This is a gift that helps me grow and improve, and will not go unappreciated in me.